Compositional Processes as Graph

Semantic Modelling and Terminology in the Anton Webern Gesamtausgabe (AWG)

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Question. How can a digital edition represent non-linear compositional processes – and keep philological claims as open as the evidence demands?

Answer: partly. `concomitates`, plain `precedes` and a certainty tag keep claims open. Elsewhere openness survives only as free text – readable, not queryable.

1 The AWG: three forms of edition, one graph in two layers

Form of edition	Presents	Edits
work edition · Werkedition	versions published or meant for print	`TextVersion`
text edition · Textedition	versions beyond a sketch, not printed	`TextVersion`
sketch edition · Skizzenedition	diplomatic transcriptions of sketches	`Sketch`

Generated layer
derived from the edition data

Curated layer
asserted by the editors

Draft ontology v0.1 · SPARQL in the app · Turtle export · no reasoning

2 Op. 25: the sketches cross the opus order

Op. 25 = M 317 (song I) · M 322 (song II) · M 321 (song III). The curated chain:

M 317 Sk4 → **M 321 Sk1–9** → **M 322 Sk1**

Song III was sketched before song II. The generated layer shows three separate “islands”.

3 Terms from the material become classes

Term	In the documents	Class
sketch · Skizze	Sk1, Sk2, Sk2.1	`Sketch ⊂ Inscription`
row table · Reihentabelle	Webern’s sheet of row forms	`RowTable ⊂ Sketch`
paratext · Paratext	title, dating, instruments	`Paratext ⊂ Inscription`
version · Textfassung	TF1, TF2	`TextVersion`
edition complex · Editionskomplex	opus or M number	`EditionComplex`
uncertainty	“presumably”	tag `certainty: uncertain`

Not modelled yet: “study composition”, “ink draft”.

4 Two relations carry the genetic order

Relation	Meaning	cf. CIDOC CRM
`precedes`	A begun before B, overlap allowed	P176 “starts before the start of”
`concomitates`	A and B shape each other, no order	P174 “starts before the end of”
`precedesScripture`	evidence: the writing	
`precedesContent`	evidence: the musical content	

`precedes` is transitive; `concomitates` is symmetric, not transitive.

5 Standards serve as alignments, not as the model

A sketch's genetic place rests on content and material traces together – which LRMoo separates (F2 Expression vs. F5 Item).

Standard	In the AWG model	Link
LRMoo	edition complex ≈ F1 · version, sketch ≈ F2 · document ≈ F5	`rdfs:seeAlso`
CIDOC CRM	temporal reading: `precedes` ≈ P176 · `concomitates` ≈ P174	`rdfs:seeAlso`
DC terms, SKOS, PROV, `rdf:Statement`	parts · concept schemes · attribution · statements	used directly
MEI	not integrated yet	–

6 The model could transfer; the pipeline cannot

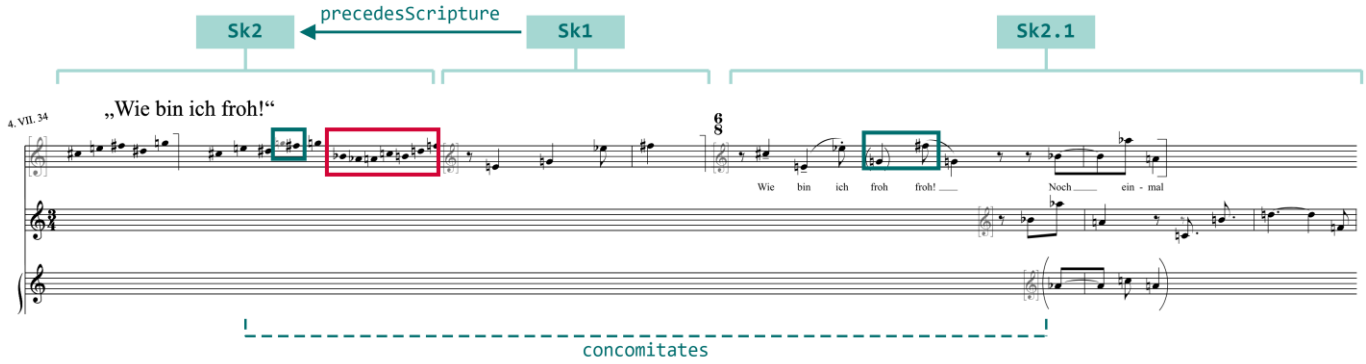
Portable

- `Inscription` · `Document` · `Folio` · `Placement`
- `precedes` · `concomitates`

Tied to the AWG

- the generator reads the AWG's own data
- Sk1 before Sk2 is read as `precedes`

7 Example: M 317, sketchbook 3, fol. 38v



Sk1 `precedesScripture` Sk2
The end of Sk2 is squeezed (red): Sk1 was there first.

Sk2.1 `concomitates` Sk2
In both, g gives way to f♯ (green): neither comes first.

Source: AWG online edition, M 317 Sk2, Sk1, Sk2.1, superimposed (development version, 12 Nov 2026).

8 What the graph shows

→ Clusters
Sk2.1 and its variants: one work phase at a glance

→ Paths
Sk1 → Sk4 → M 321: the work leaves the song

→ Gaps
a "presumably" without a tag counts as certain

9 Where model and data disagree

Model lags data

- staves, measures as free text
- no writing instruments

Data lags model

- 29 Sk## placeholders
- datings without year

12 generator warnings – each gap is information.

10 Conclusion

- **Layers** show where editorial judgment becomes explicit.
- **Warnings** show where the data disagree.
- **Model and data** are written side by side: concomitant.

Next: named graphs for ink and pencil layers · persons · archives

For discussion. Whose genetic order does the graph show – Webern's or ours?
How much of Webern's compositional process survives translation into formal models?